Younes Mouhib

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EDUCATION

08.2026 — **MSc in Mathematics** — ETH Zurich
Courses: Network and Integer Optimization, Graph Theory, Structural Graph Theory, Algebraic Methods in Combinatorics, Algebraic Topology, Algebraic Geometry, Number Theory

07.2023 — **BSc in Mathematics** — EPFL
Courses: Probability, Statistics, Stochastic Processes, Numerical Analysis, Measure and Integration, Ergodic Theory, Functional Analysis, Discrete Optimization

RESEARCH

07.2026 — **Master thesis** (Prof. Dr. Lorenz Halbeisen) — ETH Zurich
SAT-based search: a symmetry reduction turned an intractable problem with over 10 billion points into a small constraint model (253 variables, 1,771 constraints) solved in seconds by CryptoMiniSat, setting new record lower bounds for the Hales–Jewett numbers, independently verified by code (GitHub).

06.2026 — **Preprint** — arXiv:2606.22155
New records HJ(3,3) from 14 to 22 and HJ(4,2) from 12 to 14.

09.2025 – 12.2025 — **Student seminar in combinatorics** (Prof. Dr. Benny Sudakov) — ETH Zurich
«The Graph Regularity Method», two-hour talk

07.2023 — **Bachelor thesis** (Prof. Dr. Maria Colombo, Dr. Lucio Galeati) — EPFL
«On the Subadditive Ergodic Theorem and Its Applications»

PROJECTS

Machine learning — **Sudoku vision** — GitHub
A convolutional network (PyTorch) reads handwritten sudoku grids, a SAT solver solves them and MaxSAT repairs misread digits. Fully correct grids rose from 75% to 99% on 1,000 test puzzles.

AI agents — **Brigade, multi-agent LLM prover** — GitHub
LLM agents plan and write mathematical proofs, and only the Lean 4 proof checker can accept them. FastAPI backend, Docker image, continuous integration.

EXPERIENCE

06.2026 — **Course instructor, Topology (PVK)** — ETH Zurich
Designed and taught a four-day exam-preparation course in topology.

07.2021 – 01.2023 — **Homework tutor, mathematics and science** — Lausanne
Supported pupils in grades 9–11 at the Collèges Villamont and St-Roch, with individual study plans.

04.2022 — **Science outreach volunteer** — Thun
Ran hands-on science activities at the EPFL Science Day in Thun.

09.2020 – 02.2021 **Student mentor** EPFL
Mentored first-year students during the COVID-19 pandemic.

Part-time jobs to finance my studies Lausanne and Zurich
Warehouse and logistics (FMEL), waiter (Tribeca, Le Cailhau), private tutoring in mathematics.

SKILLS

Programming	Python, SQL, MATLAB, Lean 4
Data and ML	NumPy, pandas, scikit-learn, PyTorch
Optimization	Gurobi, SAT and MaxSAT solvers (PySAT, CryptoMiniSat)
Tools	Git, continuous integration, Docker, Linux, LaTeX, Excel

LANGUAGES

French	Native
Moroccan Arabic	Native (Darija)
English	C1
German	A1
Spanish	B2
Indonesian	A2

FURTHER EDUCATION

11.2026 **SQL for Data Science,** University of California, Davis Coursera

11.2026 **Google Cybersecurity Professional Certificate,** Google Coursera

INTERESTS

Chess, literature, martial arts, guitar; occasional member of a debate club and a chess club

REFERENCES

Available upon request